

Connected non-flavor advances

Native arithmetic proofs, bounded streams and certified marked monodromy

Complete native rational roots and elliptic two-torsion, compositional finite-domain certificates, noncoprime residue charts, variable-width factorial obstructions, rectangular determinants, exact graph metrics and rational analytic enclosures.

42

audited public Lean declarations

277,345

exact SMT command slices
compared

148

certified steps around a
Legendre loop

New mathematical proofs and bounded executable extensions. Historic census proofs retain their own receipts; open research premises and unverified interpreter boundaries remain explicit.

Source, corpus receipts and reproduction commands accompany the repository snapshot.

The delivered boundary

This edition records the 6 October 2026 continuation of the elliptic-division push. It works on the arithmetic, compiler, curve, analytic, geometry, graph and application fronts.

Flavor is reserved for a separate session. The new mathematical layer has seven Lean modules and 42 explicitly audited public declarations, plus one separately audited generated factorial theorem. The chapter distinguishes a proved mathematical reduction, an executable producer, a source-bound replay and a still-open global premise.

Completing the new reductions does not discharge every research problem mentioned in the earlier frontier review.

The reproducible entry points are `make nonflavor-frontier-lean`, `make nonflavor-frontier-receipts` and `make test`. The focused Lean target builds the seven modules, checks concrete fixtures, prints each audited declaration's axioms and runs namespace lint. The generated factorial theorem has its own axiom check. Only the standard axioms `propext`, `Classical.choice` and `Quot.sound` are permitted. The historical whole-repository Lean census is not rebuilt by this focused target; its old counts retain their original verification scope. The complete archive includes every tracked repository path.

Native roots become actual elliptic fibres

`NativeRationalRoots` turns the proved signed-divisor integer generator into a complete rational-root generator. A producer supplies an ascending integral coefficient list, a nonzero rational scale D , monicity and an exact coefficient identity with the scaled polynomial. Lean checks those premises. The integer-root output is obtained from `NativePolynomialRoots`; it is no longer supplied as a separately assumed complete list. Roots are transported back by z/D . A leading-coefficient normalization extends the result to every nonzero rational polynomial for which the checked scaled integral coefficients are supplied. Zero constant terms and repeated roots retain the existing recursive treatment.

`EllipticPointDivision` connects the completed-square model $y^2 = x^3 + Ax^2 + Bx + C$ to Mathlib's actual affine point group. For nonsingular affine points and nonzero 2, two-torsion is equivalent to $y=0$; the x coordinate is therefore a cubic root. The x coordinate in Mathlib's doubling formula satisfies exactly the halving quartic proved in the previous push. A nonzero discriminant allows each rational cubic root to construct a nonsingular branch point. The torsion list includes infinity. Native rational roots then prove this list complete, and an actual checked anchor H with $2H=P$ gives all halves by translating that torsion list. The resulting theorem covers the entire fibre, including infinity and exceptional points through the group theorem; only the coordinate doubling equivalence uses its explicit nonzero-denominator premise.

The remaining elliptic obligations are now sharply located. The Python point-law implementation, JSON parser and certificate interpreter still need a formal refinement to these native definitions. Anchor discovery remains a producer task; its group equation is a checked premise, not a rank assertion. The independence machinery still proves the stated lower bounds under its own certificate semantics; full Mordell-Weil rank

determination, arbitrary n-division and generic higher-genus covering maps remain separate work. Generalized-model transport must continue to preserve original-coordinate integrality.

Compositional domains, CRT and nonmonotone images

`FiniteDomainCertificate` provides a transcript language for literal finite sets, unions, intersections, decidable filters and images. Its denotation retains the source predicate, and its evaluator produces a deduplicated finite set. A structural completeness theorem relates evaluation to denotation. Rank counts distinct accepted integers strictly below a value. Strict rank growth and selected-value uniqueness show that membership plus rank identifies one value. A source-equivalence theorem transports that guarantee back to a supplied original predicate. This is useful for nonlinear images because several source parameters can have one output value; counting source occurrences would give the wrong rank.

`GeneralCRT` removes a hidden coprimality restriction from the mathematical chart layer. A simultaneous pair of natural residue conditions exists exactly when the residues agree modulo the gcd. The canonical compatible anchor has an lcm-period chart and, for nonzero moduli, lies below that lcm. A separate integer theorem keeps the same chart exact for negative parameters and residues. The remaining compiler work is a universal parser/AST correspondence, generic Sturm-variation certificates, assembly of every emitted semilinear cell and a verified algorithm for rank/select and optimizer search. The new transcript theorem proves the meaning of assembled checked data; it does not claim that an arbitrary Python compilation already constructs those proofs.

The population implementation now supports tuple-key joins. Bounded value joins preserve every original pair and every multiplicity. Symbolic joins choose one complete curve relation and enforce the additional polynomial equalities through the curve predicate language. A test joins the square and linear fields over $2 \times 10^{30} + 1$ source parameters and retains exactly the diagonal without enumeration. Unsupported primary curve relations still fail explicitly. Arbitrary multivariate joins and a general complete relation solver remain open.

Nonmonotone projection now includes integer even quartics ax^4+bx^2+c . `QuarticCollision` proves that equal values have $x=y$, $x=-y$ or $a(x^2+y^2)+b=0$, and proves bounds for the circle branch. The Python backend handles the infinite reflection through source predicates and exhausts the finite circle when its work bound permits. It retains the least original accepted owner even on disconnected domains, then reuses the existing image/rank interfaces. Arbitrary nonsymmetric quartics and higher-degree nonmonotone collision curves still require new complete backends; a budget rejection returns no partial completeness claim.

Variable-width factorial arithmetic

FactorialWindow proves the p-adic valuation of $N!/(N-W)!$ as a difference of factorial valuations and then as two finite Legendre sums. At the native level, natural subtraction makes $W > N$ a truncated window; the public Python operation requires $0 \leq W \leq N$. A valuation not divisible by k proves that the positive window product is not a k -th power. The Python witness search uses the bit length of N and a bounded supplied prime list, so it can process widths that cannot be enumerated. Passing every tested prime produces UNKNOWN.

The retained generated theorem proves that $(10^{30})!/(10^{30}-10^{20})!$ is not a square. Its 2-adic valuation is 100000000000000000011, which is odd. Lean evaluates the finite floor sums and checks logarithmic truncation bounds through a power inequality; it does not reduce either enormous factorial. This closes a useful variable-width Gamma-ratio obstruction route. Complete classification of arbitrary variable-width products, sums or differences of factorials, general Gamma arguments and multivariate creative telescoping remain open. Sequence-definition review, continued-fraction execution at very large unit height, broad nonlinear recurrences, transform endpoints and Tauberian conclusions retain their previous boundaries.

Certified periods, inverses and marked monodromy

The analytic extension uses rational complex centres and an induced row norm with $|\operatorname{Re} z| + |\operatorname{Im} z|$ on entries. For a supplied enclosure $\|A - C\| \leq e$, exact Gaussian elimination computes C^{-1} . If $q = \|C^{-1}\|e < 1$, Neumann's lemma proves nonsingularity throughout the ball and bounds inverse error by $\|C^{-1}\|q/(1-q)$. Product balls use the full cross term. Integral recognition requires error strictly below one half, checks entry and row compatibility, and can check the standard symplectic form exactly. The general operation explicitly requires an independent integrality premise: closeness alone cannot establish integrality.

Long analytic paths now have an optional outward-rounded dyadic mode. Centres are rounded with a bounded row error added to the enclosure; the error bound itself is rounded upward. Certified disks also keep the exponential bound controlled. This prevents repeated rational matrix products from accumulating huge denominators while preserving the existing exact mode. The marked Legendre wrapper combines the original certified seed, closed continuation, inverse and product balls, and symplectic recognition. Its integrality justification is the supplied standard integral cycle marking, not numerical proximity.

The retained nontrivial loop starts at $1/2$, passes through $1/2+i/4$, $-1/4+i/4$, $-1/4-i/4$ and $1/2-i/4$, and returns to $1/2$. At Taylor order 24 and 128-bit outward rounding, 148 certified steps identify the cycle matrix $[[1,2],[0,1]]$. The row enclosure is approximately $2.22466826496 \times 10^{-9}$. This is a concrete certified marked monodromy computation. General singular endpoint asymptotics, arbitrary integral period markings, rigorous quadrature with arbitrary cycle inputs, holomorphic-basis normalization and the analytic estimates themselves in Lean remain open.

Curve structure, arithmetic reduction and global geometry

The existing exact curve engine retains its explicit quotient towers, two-isogenies, Richelot constructions, de Rham connections, superelliptic reductions, horizontal-section searches and selected Frobenius computations. The current push adds native elliptic fibres and a certified marked analytic continuation bridge. General higher-genus correspondences still need integral cycle maps, kernel degrees and singular-target treatment. Nonsplit stable reduction, arbitrary Puiseux root jets, arithmetic component twists, finite-field extension descent, cross-parameter p-adic Frobenius transport, full differential Galois groups, nonlinear idempotents and conjugate Hodge/Betti descent remain research fronts. A bounded ansatz search for horizontal sections cannot prove that every possible section has been found.

`cluster_graph_metric` turns the existing split-root geometric semistable graph into a global finite length metric and tropical cycle lattice. Exact Floyd-Warshall distances respect all graph identifications and parallel edges. A deterministic spanning tree gives integral fundamental cycles; every boundary is checked to vanish, the number of cycles matches the graph's cycle rank, and the edge-length Gram form has a positive determinant. A replay operation binds the entire packet to the original rational roots and odd prime and rejects changed source data or witnesses. These distances and cycle lengths are properties of the geometric reduction graph. They are not a complex-curve metric, a minimal arithmetic regular model or a Frobenius marking.

The remaining smooth geometry work includes a global quotient length metric on the analytic atlas, JSON-to-native geometric checker correspondence, global chart gluing and smoothness comparison, general period-normalized metrics and certified geodesics. The existing 108 closed metric boxes out of 144 leave 36 unresolved boxes; this push does not turn them into certified boxes. Numeric hyperelliptic periods remain scoped to their supported genera and numerical methods, and general cyclic-mesh cycle integration still needs a marked analytic bridge.

Graph determinants and probabilistic interpretation

`RectangularDeterminant` proves the arbitrary-dimensional multilinear expansion of $\det(MN)$ as a sum over selected rows, with a determinant of the selected columns of M and a product of entries of N . Repeated selections have equal columns and vanish. The sum can therefore be restricted to injective selections, including rank-deficient shapes with fewer rows than columns. This removes the two-column restriction at the determinant-expansion stage and provides the algebraic foundation for a general Cauchy-Binet formalization.

Grouping these signed ordered selections into unordered weighted squared minors remains a separate proof. So do the general gain-graph support/rank correspondence, matroid greedy interpretation and a formal algebraic-bit sampler. The existing finite probability-law and exact conditional-measure implementation keep their current guarantees. A passing Python determinant or sampling replay does not by itself establish an end-to-end probabilistic implementation refinement in Lean.

Solver transport and verification consumers

`iter_commands` accepts bounded text chunks and yields exact SMT-LIB command slices on demand. Its state handles comments, quoted symbols and doubled string quotes across every chunk boundary. It rejects malformed tails, top-level garbage and exceeded command or chunk limits. `solve_replay_stream` preserves native query barriers and batches only non-query transport. A query sink avoids retaining rows. Errors stop future queries while preserving completed earlier query results. Storage bounds cover the scanner and pending transport; they do not bound the solver's assertion state, the caller's input storage or an intentionally retained result list.

The scanner matched 277345 exact slices across 292 repository SMT files at a 127-character chunk size. A source-hashed replay of 16 repository and Why3 tasks reported no sat/unsat contradictions. Scanner allocation measurements used generated chunks and measured 1332 bytes for 1000 commands and 1340 bytes for 100000 commands. They are local measurements, not whole-process memory bounds. The earlier external 181-file industrial cohort is not replicated by this receipt. Its reported throughput and solved-query gains retain their earlier provenance, and independent replication and broader throughput work remain necessary.

The Why3 bridge now allocates distinct names after sanitization, reserves generated local names and language keywords, safely encodes source-symbol comments, caps expanded powers and rejects unsupported free Boolean binders. These fixes eliminate concrete malformed or capturing translations. The original Why3 tool is unavailable in this environment, so its native rerun remains skipped. A universal SMT-to-Why3 semantic correspondence, SPARK/GNATprove VC correspondence, overflow and bitwise refinements remain open. Brainpool's residue-ring equation/export transport remains proved, while its 384-bit field primality and complete group implementation bridge still require a native prime certificate; trial factoring of $p-1$ leaves a composite 349-bit cofactor.

Global arithmetic and next research decisions

The historical descent coverage still records 79 classes: 15 local and 64 conditional, with zero unresolved coverage entries. Unit registration covers every currently needed obligation; this does not discharge the global Matveev-type premises. The 23 negative curves carrying those premises retain conditional theorem statements. The positive registry still records 73 closed cases and the earlier conditional entries. $k=22$ remains closed by its retained 100-chunk kernel check of 971659 source candidates. $k=94$ remains the next larger source with 5294219 candidates; no complete source theorem for it is claimed by this push. Generic native maximal-order computation, basis-independent nonmonic norm transport and recentered Skolem extraction remain separate from the rational-polynomial normalization delivered here.

Density-zero statements for general nonrigid branches, normalized Kummer/Riemann-Hurwitz statements and global Siegel/Boshernitzan ingredients remain mathematical obligations. Hall, Pillai and uniform integral-point conjectures remain

outside the unconditional results. The next productive work is to connect the new native elliptic and finite-domain definitions to emitted certificates, complete the unordered weighted determinant identity, extend certified marked periods to singular and higher-genus settings, and choose an arithmetic closure with an affordable source-proof budget. No roadmap item is promoted to a theorem merely because it now has a working data pipeline.